

Question Paper

Exam Date & Time: 09-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JUNE 2025

SUBJECT: CSE 2222/CSE_2252 - DESIGN AND ANALYSIS OF ALGORITHMS

(SPL: COMPUTER SCIENCE AND ENGINEERING - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING / COMPUTER SCIENCE / COMPUTER SCIENCE AND ENGINEERING - CYBER SECURITY)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) You are developing a smart attendance system for a university. Each class session has a list of student entry times. A student is considered "on time" if they arrive within 5 minutes of the session start time (e.g., session starts at 09:00, on-time arrival is up to 09:05). Develop an algorithm to count how many students arrived on time. The list of arrival times is provided in HH:MM format. (3)

- 1B) Consider the following non-recursive algorithm: (3)

Algorithm Example1(A[1...n])

sum = 0

for i = 1 to n do

for j = 1 to i do

*sum = sum + A[i] * A[j]*

return sum

Analyze the time complexity of this algorithm and express it using Big-O notation, showing the steps of your derivation.

- 1C) Determine the asymptotic time complexity of the recurrence relations: (4)

i) $T(n) = 2T(n/2) + n$

ii) $T(n) = 4T(n/2) + n^2$

- 2A) Consider a sorted array with duplicate elements, e.g., $arr = [2, 5, 5, 5, 8, 12]$. If we use the standard Binary Search code to search for $target = 5$, which index (if any) is guaranteed to be returned? Explain why. How could you modify the Binary Search algorithm to find the first occurrence of the target in such a case? Write an algorithm for this modified approach. (4)

- 2B) Explain the basic principle of the Brute-Force String Matching algorithm. Illustrate its operation by manually tracing the algorithm to find the pattern **ABA** in the text **CABABABA**. Show the character comparisons made at each step. (3)

- 2C) How does the performance of Sequential Search change if the array being searched is sorted? Does sorting the array improve the worst-case time complexity of a standard Sequential Search? Explain your reasoning with one better algorithm and write the algorithm. (3)

- 3A) Analyse the worst-case time complexity of Merge sort algorithm considering key comparison as the basic operation. Apply the Merge sort algorithm to the following list to arrange in ascending order: 70, 60, 50, 40, 30, 20, 10 and find the total number of key comparisons (3)

- 3B) What is 2-3 tree? Construct 2-3 tree for the list **U,N,C,O,P,Y,R,I,G,H,T,A,B,L,E** by successive insertion method showing all stages (3)

3C) Sort the list: **A, L, G, O, R, I, T, H, M** in alphabetical order using heapsort by clearly showing bottom-up heap construction and sorting stages. (4)

4A) Solve the Knapsack problem using dynamic programming with memory functions (top-down memorization) for the data given in Table 4A, with capacity $W = 7$. (4)

Table 4A

Item	Weight	Values
1	2	30
2	3	50
3	4	60
4	5	70

4B) Given the following keys: "greedy", "divide", "conquer", "dynamic", "search", "graph", "tree", "recursion", and "backtrack" construct a hash table using open hashing (chaining). Use the hash function $h(\text{keys}) = (\text{alphabet sum}) \bmod 11$ (4)

i) Construct the hash table.

ii) What is the load factor of the hash table?

4C) Apply Warshall's algorithm to find the transitive closure of the given graph shown in Figure 4C. (2)

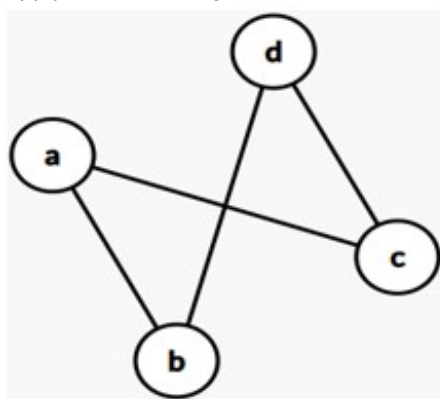


Figure 4C.

5A) Apply Prim's algorithm to the following graph shown in Figure 5A. Include in the priority queue only the fringe vertices. (4)

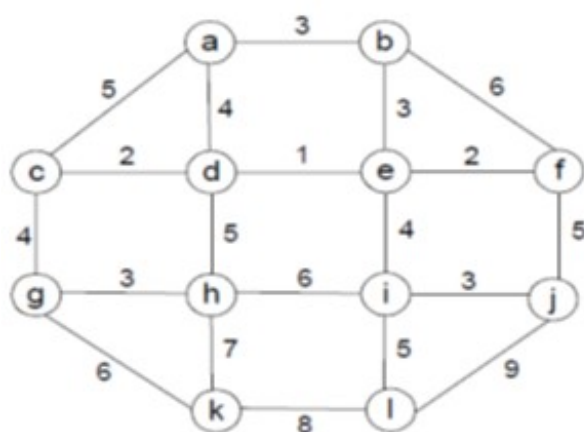


Figure 5A

5B) Draw the complete state-space tree of the backtracking algorithm applied to the instance $A = \{3, 5, 7, 9\}$ and $d=12$ of the subset sum problem. (3)

5C) Solve the assignment problem using the Branch and Bound technique for the given 4x4 cost matrix representing the cost of assigning each machine (M1, M2, M3, M4) to each job (J1, J2, J3, J4). Determine the optimal assignment that minimizes the total cost and draw the state-space tree. (3)

	J1	J2	J3	J4
M1	11	12	18	40
M2	14	15	13	22
M3	11	17	19	23
M4	17	14	20	28

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Event Name	Subject
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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025 CSE-2252:Design and Analysis of Algorithms

#	Course Outcome	Description	Marks
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No records found / CO Level Description not defined

* **NA** - CO Not Selected

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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025 CSE-2252:Design and Analysis of Algorithms

#	Level	No of Questions	Marks (%)
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